CUSTOMER MILESTONE BRIEF

MS #18 - CICS, VSAM, IMS, and HLASM Readiness Proofs

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.18.0, 0.18.1, 0.18.2, 0.18.3, 0.18.4, 0.18.5
Date	2026-08-21 to 2026-08-22
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #18 - CICS, VSAM, IMS, and HLASM Readiness Proofs converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Establishes bounded logical proof cells for key mainframe technologies before the later full qualification milestones.

Why this matters to a customer

Establishes bounded logical proof cells for key mainframe technologies before the later full qualification milestones.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added deterministic native extraction for CICS CSD transactions, programs, files and mapsets; BMS maps and fields; EXEC CICS command spans; and IDCAMS VSAM clusters, components, alternate indexes and paths, each with exact source evidence.
- Added typed routing and lineage relationships from CAVW through COACTVWC, CACTVWA, CICS file resources, and the underlying CardXref, account, and customer VSAM objects.
- Curated eight graph-grounded account-view behavior rules and a bounded read-only modernization candidate with an independent private gate and mutation/negative coverage.
- Added an operator-safe real-CICS capture contract, redacted artifact manifest, mainframe identity requirements, differential comparator, and fail-closed signed readiness receipt.
- Added cross-platform launchers, JSON schemas, an operational runbook, deterministic development evidence, and full verification integration. Local proof cannot satisfy z/OS equivalence.
- Added deterministic HLASM parsing for programs, instructions, symbols, branches, macros, DSECTs, and fields.

- Added native IMS DBD/PSB parsing for databases, dataset groups, segments, fields, PCBs, sensitive segments, and program-to-PSB bindings.
- Added a bounded, source-faithful COBDATFT modernization candidate with private mutation gates.
- Added a graph-bound CICS, VSAM, IMS, and HLASM capability projection across readiness gates 1-8.
- Kept every technology fail-closed for mainframe equivalence until signed z/OS evidence exists.
- Added a bounded logical proof for the CBPAUP0C IMS BMP authorization-expiry workload through PSBPAUTB, PAUTBPCB, DBPAUTP0, PAUTSUM0, and PAUTDTL1.
- Curated eight source-grounded rules for BMP routing, hierarchy, GN/GNP traversal, inverted-date expiry, summary adjustments, DLET behavior, checkpoints, and the duplicated approved-count root deletion quirk.
- Added a source-faithful candidate, independent private mutation gate, deterministic local capture, differential comparator, z/OS capture contract, operational runbook, and signed IMS readiness receipt.
- Advanced IMS readiness gates 3-5 to passed and gate 7 to mechanism-ready while keeping live BMP execution and mainframe equivalence blocked until externally attested z/OS evidence exists.
- Added one adaptive PowerShell Python runtime resolver across every Windows entry point, with an explicit LIGHTYEAR_PYTHON override and tested support for Python 3.11 and newer.
- Made managed AWS CardDemo checkouts line-ending neutral and added repository-wide Git attributes for stable text and binary handling.
- Added dual source identity: raw transport hashes preserve forensic custody while normalized-LF logical hashes drive graph and evidence-pack semantic identity.
- Made canonical JSON and Markdown writers emit UTF-8/LF on every platform.
- Added Windows and Linux CI plus regression tests for LF/CRLF equivalence, canonical receipt identity, shared launcher adoption, and executable shell entry points.
- Replaced dictionary-only comparator indexing with explicit duplicate detection on both expected and actual records, plus independent population-count checks.
- Added a three-state differential contract: equivalent exits 0, verified differences exit 1, and comparisons with no evidence return indeterminate and exit 2.
- Removed timestamp suppression because the oracle and candidate already receive the same pinned clock; malformed, blank, or different timestamps now fail comparison.
- Added a versioned, content-addressed comparison report and an owned normalization ledger.
- Made baseline_first, allow_network, and expose_output_to_builder reject quoted or numeric pseudo-booleans; deserialized reports require an exact true before exposing gate output.
- Added comparator escape and holdout-boundary suites with positive controls, bare-pytest source discovery, cross-platform verifier launchers, dedicated CI jobs, and a tracked-evidence clean-tree assertion.
- Kept the separately observed oracle duplicate-account/CardXref behavior change out of scope until it receives an independent source and z/OS semantics review.

- Pinned the direct-construction default for private gate output so a future dataclass-default inversion is rejected even when no JSON deserialization path is involved.
- Pinned the comparator's first-observed duplicate diagnostic policy while preserving independent duplicate and population-count failures.
- Turned the normalization ledger into an executable governance control: schema, comparator, runtime scope, behavior, owner, reason, and ISO review date are validated fail-closed.
- Made a normalization review date expire on the stated date and wired validation into both the focused verifier gauntlet and full cross-platform verification.
- Clarified that the tracked-evidence clean-tree assertion is a preventive CI control; it was not remediation of an independently observed artifact-mutation defect.

How it differs from earlier work

MS #18 builds on MS #17 (Live Evidence Control Tower). The earlier milestone established its own bounded capability; this milestone adds cics, vsam, ims, and hlasm readiness proofs while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use cics, vsam, ims, and hlasm readiness proofs as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Added cross-platform launchers, JSON schemas, an operational runbook, deterministic development evidence, and full verification integration. Local proof cannot satisfy z/OS equivalence.
- Kept every technology fail-closed for mainframe equivalence until signed z/OS evidence exists.

Source of record

- CHANGELOG.md release section(s): 0.18.0, 0.18.1, 0.18.2, 0.18.3, 0.18.4, 0.18.5
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications

- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity